

BaseCenter.ai — Primary System Build Document (Phase 1)

Document Version: 1.1.0

Status: Approved Architecture

Target Platform: Abacus.ai Cloud Infrastructure / Next.js / FastAPI / PostgreSQL

Document Purpose: Master blueprint for all autonomous development agents and engineers building the BaseCenter.ai platform.

1. Executive Summary & Core Business Logic

1.1 Product Vision

BaseCenter.ai is an **AI-customizable business operating system** designed for solo entrepreneurs, small businesses, contractors/trades, and growing enterprises. The platform is built on the core promise:

“Business software that adapts to how you work — not the other way around.”

1.2 Commercial & Packaging Engine

- **10 Core Paid Business Modules + 1 Optional Website Module:**

1. Invoice & Milestones (SOW generator, milestone tracking, automated billing, payment gateways)
2. Contracting (AI legal document generation, redlining, approval workflows)
3. Help Ticket (AI-assisted customer & internal support desk)
4. Knowledge Base (Dynamic context-aware documentation builder)
5. Data Collection (Custom form builder with prompt-based processing pipelines)
6. Project Management (Hybrid Kanban/Scrum with autonomous AI coordination)
7. File Management (Cloud document suite with Google Docs / Dropbox parity)
8. Training LMS (Autonomous course curriculum generator & scheduler)
9. Accounting (Configurable ledger, invoicing sync, QuickBooks/Intuit export)
10. CRM Plus (Customer lifecycle organizer, lead scoring, automated campaign engine)
11. Website Module (Drag-and-drop block website builder, 10 business themes, free inclusion)

- **The “1 Free + Free Website” Acquisition Model:**

- Every tenant selects **1 Core Business Module 100% Free** (5-year guaranteed introductory period).

- The **Website Module is included at \$0** with every account (optional activation).

- The free module tier includes **10 full-featured user seats**.

- **Monetization & Add-On Unit Economics:**

- Additional modules cost **\$5.00 / month per module**.

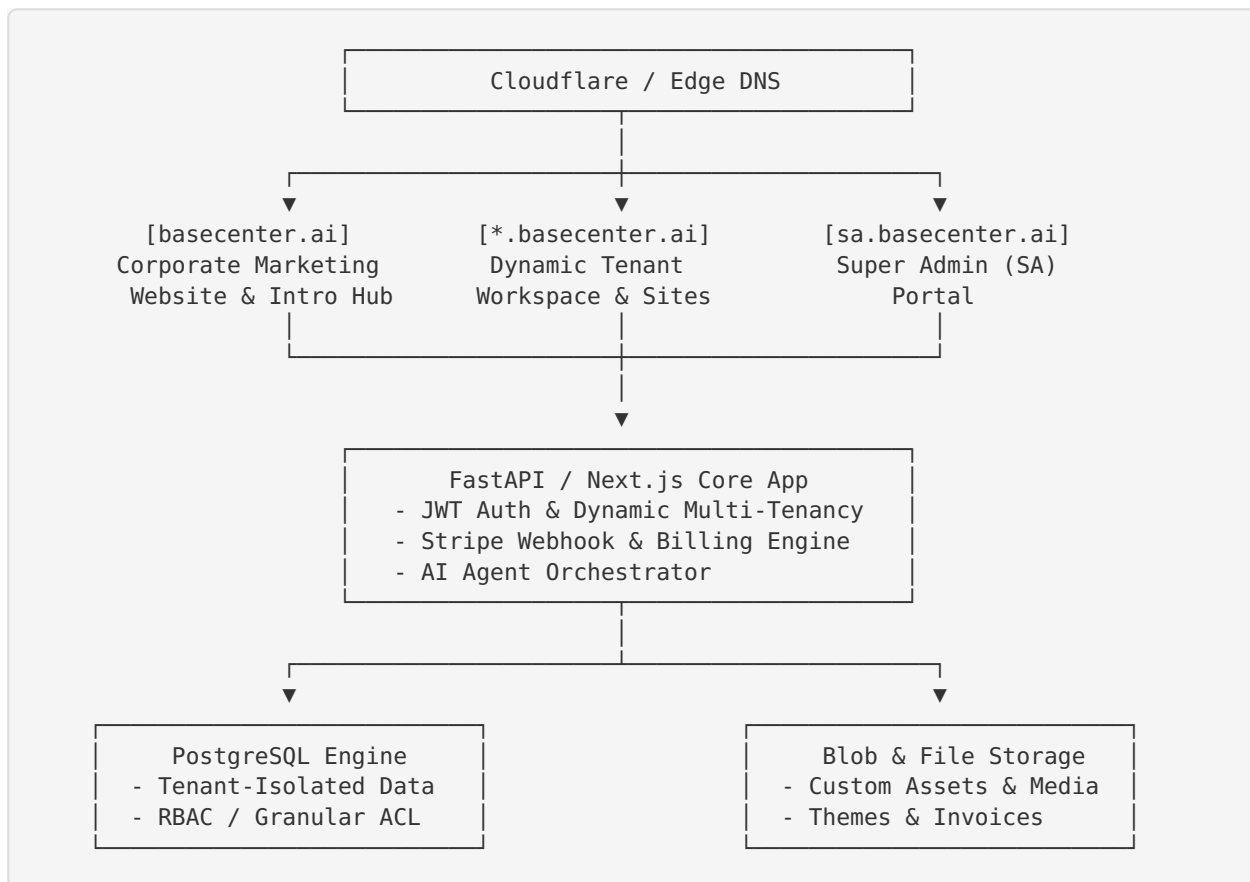
- Each activated module includes **10 seats**.

- Additional seat expansion: **+\$5.00 / month per 10 additional seats**.

- Free Module Switch Policy: A customer may switch their free module **exactly once** via an automated token/coupon emailed to the Account POC.
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2. Multi-Tenant Architecture & Routing Topology

2.1 Dynamic Subdomain & Edge Routing



- `basecenter.ai` / `www.basecenter.ai` : Corporate public marketing site, 10-module interactive selector, pricing calculator, and auth entry.
- `app.basecenter.ai` / `[tenant].basecenter.ai/app` : Authenticated tenant workspace console.
- `[tenant].basecenter.ai` : Public-facing tenant marketing website (rendered dynamically if the Website module is enabled).
- `sa.basecenter.ai` : Role-gated Super Admin control center.

3. Data Schemas & Access Control Layer (PostgreSQL)

```
-- Schema: Core Multi-Tenant Structure
```

```
CREATE TABLE organizations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(255) NOT NULL,
  slug VARCHAR(64) UNIQUE NOT NULL, -- e.g., 'acme' -> acme.basecenter.ai
  status VARCHAR(32) DEFAULT 'pending_onboarding', -- 'pending_onboarding', 'active', 'delinquent', 'suspended'
  stripe_customer_id VARCHAR(128) UNIQUE,
  stripe_payment_method_id VARCHAR(128),
  billing_status VARCHAR(32) DEFAULT 'good_standing',
  has_website_activated BOOLEAN DEFAULT FALSE,
  website_domain VARCHAR(255),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

```
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id UUID REFERENCES organizations(id) ON DELETE CASCADE,
  email VARCHAR(255) UNIQUE NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  full_name VARCHAR(255) NOT NULL,
  role VARCHAR(32) NOT NULL, -- 'super_admin', 'staff_admin', 'account_poc', 'account_admin', 'module_manager', 'user', 'public'
  is_poc BOOLEAN DEFAULT FALSE,
  status VARCHAR(32) DEFAULT 'active',
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

```
CREATE TABLE organization_modules (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,
  module_code VARCHAR(32) NOT NULL, -- 'invoicing', 'contracting', 'help_ticket', 'knowledge_base', 'data_collection', 'project_mgmt', 'file_mgmt', 'lms', 'accounting', 'crm_plus', 'website'
  is_free_tier BOOLEAN DEFAULT FALSE,
  active BOOLEAN DEFAULT TRUE,
  seat_limit INT DEFAULT 10,
  additional_seats INT DEFAULT 0,
  stripe_subscription_item_id VARCHAR(128),
  subscribed_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
  CONSTRAINT unique_org_module UNIQUE (org_id, module_code)
);
```

```
CREATE TABLE free_module_switches (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,
  original_module_code VARCHAR(32) NOT NULL,
  new_module_code VARCHAR(32),
  coupon_code VARCHAR(64) UNIQUE NOT NULL,
  is_redeemed BOOLEAN DEFAULT FALSE,
  redeemed_at TIMESTAMP WITH TIME ZONE,
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

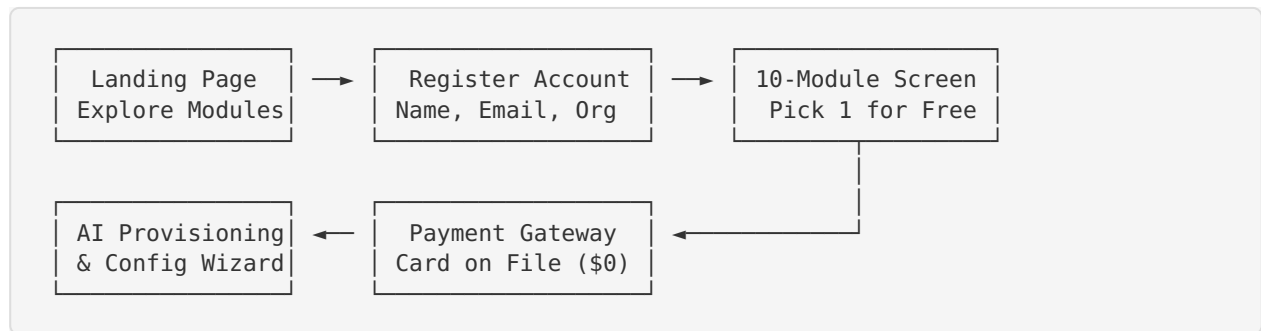
```
CREATE TABLE custom_access_control (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,
  role_name VARCHAR(64) NOT NULL,
  module_code VARCHAR(32) NOT NULL,
  can_create BOOLEAN DEFAULT FALSE,
```

```

can_read BOOLEAN DEFAULT TRUE,
can_update BOOLEAN DEFAULT FALSE,
can_delete BOOLEAN DEFAULT FALSE,
field_level_restrictions JSONB DEFAULT '{} '::jsonb
);

```

4. End-to-End Customer Onboarding & Provisioning Pipeline



1. Step 1: Registration Form

- * Fields: `full_name`, `email`, `password`, `organization_name`.
- * Organization record created with `status = 'pending_onboarding'`. No cloud resources provisioned yet.

2. Step 2: Interactive 10-Module Grid

- * High-contrast cards for all 10 modules + Website module toggle.
- * Customer selects their 1 Free Core Module + optional Free Website Module.

3. Step 3: Secure Payment Details Capture (\$0 Upfront)

- * Stripe Elements `SetupIntent` collects credit/debit card on file.
- * Clear Reassurance Banner: "Your first module is free for 5 years with 10 seats included. You will not be charged today."

4. Step 4: Dynamic Tenant Provisioning & AI Wizard Execution

- * Tenant status transitions to `active`.
- * Record inserted into `organization_modules` (`is_free_tier = TRUE`).
- * AI Module Customization Agent initializes interview session to adapt data fields, workflows, and templates.

5. Super Admin (SA) Portal Architecture

Accessible strictly at `sa.basecenter.ai` for `super_admin` and `staff_admin` roles.

5.1 The 5 Core Management Engines

1. CRM (Tenant & Organization Management)

- * Multi-tenant directory with filters for active modules, seat usage, and payment standing.
- * Secure, audit-logged **Tenant Impersonation ("Login-As")** engine.
- * Organization lifecycle controls (`Active`, `Delinquent`, `Suspended`, `Archived`).

2. Analytics (Platform Metrics & Intelligence)

- * Real-time MRR, Average Revenue Per Account (ARPA), and cohort retention.

- * Module distribution heatmap (most popular free selections vs. top \$5 add-ons).

- * AI token consumption and compute metrics per tenant.

3. **Billing/Accounting (Subscription & Payment Hub)**

- * Stripe charge logs, dunning retries, and invoice reconciliation.

- * Coupon & Switch Token Generator for 1-time free module transfers.

- * Manual credit issuance, seat overrides, and refund tools.

4. **System Settings (Global Configurations & AI Prompts)**

- * Master React row/section website editor controls.

- * Global library manager for the 10 pre-built website themes.

- * System Prompt Engineering Console for cross-module AI agents.

5. **Security (Staff Administration, ACL & Audit Trails)**

- * Staff credential manager with mandatory MFA enforcement.

- * Platform-wide immutable audit trail (logs actor, IP, timestamp, target entity, and diff).

6. Design & UI Reference Framework

Note: These SaaS applications represent the general style and theme for BaseCenter.ai. The focus is on **in-app interface design patterns**, not marketing websites.

6.1 Design System Inspirations

Monday.com — Vibe Design System

- **Core Philosophy:** Open-source React component library built for consistency and native platform feel.
- **Key Patterns to Adopt:**
 - Comprehensive design token system (colors, shadows, dimensions, typography)
 - 50+ reusable components (buttons, popovers, navigation, inputs)
 - Responsive action feedback (0.5-1s response times with loading indicators)
 - Native integration patterns for settings, search bars, and filters
 - Tooltips, hints, and banners for contextual assistance
 - Welcome pages and onboarding for first-time module activation
 - Light and dark mode support
- **Technical Resources:**
 - Vibe Design System: <https://vibe.monday.com/>
 - Component Library: `@vibe/core`
 - Figma UI Kits available for prototyping
- **Implementation for BaseCenter.ai:**
 - Build a similar React component library for cross-module consistency
 - Ensure all modules share consistent interaction patterns
 - Implement comprehensive onboarding flows for each activated module

Trello — Kanban & Visual Task Management

- **Core Philosophy:** Clean, column-based layout with intuitive drag-and-drop for visual workflow tracking.
- **Key Patterns to Adopt:**
 - Horizontal scrollable board layout with vertical columns (lists)

- Drag-and-drop interactivity with visual feedback (shadows, tilting during drag)
- Card-based content containers with hover states for secondary actions
- Contextual modals for detailed task views without leaving board context
- Real-time collaboration with presence indicators
- **Technical Implementation:**
 - Use libraries like `@dnd-kit/core` for drag-and-drop
 - Card components with metadata (avatars, comments, reactions)
 - Minimal aesthetic with hover-revealed actions
- **Application to BaseCenter.ai:**
 - Direct application to **Project Management module** (Kanban boards)
 - Card-based UI patterns applicable to **Help Ticket**, **CRM Plus**, and **Data Collection** modules

Ramp.com — Editorial Minimalism

- **Core Philosophy:** “Black-and-white editorial” design with a single high-contrast accent color for financial actions.
- **Key Patterns to Adopt:**
 - Near-monochrome palette with strategic accent color (`#e4f222` highlighter yellow)
- **Color System:**
 - Primary Accent: Highlighter Yellow (`#e4f222`) for primary actions, live counters, active states
 - Neutrals: Ink (`#0c0a08`) for primary text, Obsidian (`#1a1919`) for inverted panels, Hairline (`#e5e7eb`) for borders
- **Typography:** Single typeface (Lausanne at 400 weight), hierarchy through size and tracking
- **Elevation:** 1px hairline borders instead of box-shadows for card surfaces
- **Component Library (“Ryu”):**
 - Semantic props (e.g., `color='destructive'` or `color='constructive'`)
 - Constraint-based design to maintain uniformity
 - Isolated component library decoupled from business logic
- **Layout:** Comfortable density with 4px base spacing unit, left-aligned within centered max-width (1200px) containers
- **Border Radii:** 6px (buttons/tags), 10px (inputs), 12px (wash cards), 16px (content cards)
- **Application to BaseCenter.ai:**
 - Apply to **Accounting** and **Invoice & Milestones** modules for financial clarity
 - Use highlighter accent color for “money-moving” actions
 - Build semantic component library with constraint-based design

Attio.com — Modern Data-Driven CRM

- **Core Philosophy:** Flexible, data-driven interface with clean aesthetics and high information density.
- **Key Patterns to Adopt:**
- **Color Palette:** Signature Teal (`#3ABDAF`) for primary actions and focus states
- **Typography:** Inter typeface for high legibility
- **Nature Palette:** Green tones (Mint to Sage to Hunter) for data categorization
- **Core UI Components:**
 - Sophisticated table grids with sortable columns, custom filters, pagination, bulk actions
 - Side panels and modals to maintain workflow context
 - Well-designed empty states with zero-data screens and first-use prompts

- Modular dashboards for analytics and metrics
- Interactive onboarding with progress indicators
- Extensive form and input patterns optimized for rapid data entry
- **Application to BaseCenter.ai:**
- Direct application to **CRM Plus** module
- Table/grid patterns applicable to **Accounting**, **Data Collection**, and **Help Ticket** modules
- Side panel pattern for cross-module navigation consistency

QuickBooks Online — Professional Accounting Dashboard

- **Core Philosophy:** Clean, intuitive, modular dashboard with customizable widgets for financial oversight.
- **Key Patterns to Adopt (App Interface Only):**
- **Widget-Based Dashboard:** Customizable layout with relocatable, add/remove widgets
- **Core Functionality Tabs:** Home view for shortcuts, specialized views for cash flow, reports
- **Visual Design:**
 - High-contrast visuals with clean, legible typography
 - Grid-based compositions with modular layouts
 - Light and dark mode support (dark mode optimized for reduced eye strain)
- **Navigation Patterns:**
 - Streamlined menus with reduced steps to access tools
 - Smart search with custom filters for transactions, accounts, reports
 - Bookmarking for frequently used actions
- **Mobile Consistency:** Mirror desktop experience on mobile with task-focused interfaces
- **Note:** We are **NOT** adopting the green color from the QuickBooks marketing website. Focus is on the clean dashboard UI patterns within the app.
- **Application to BaseCenter.ai:**
- Primary reference for **Accounting** module dashboard
- Widget customization pattern applicable to main **Dashboard** across all modules
- Smart search and bookmarking patterns for platform-wide navigation

6.2 BaseCenter.ai Design System Synthesis

Based on the above references, the BaseCenter.ai design system will incorporate:

Color Strategy:

- * Primary Accent: Teal (#3ABDAF) inspired by Attio for primary actions and focus states
- * Financial Actions: Highlighter Yellow (#e4f222) inspired by Ramp for “money-moving” operations in Accounting and Invoicing modules
- * Neutrals: High-contrast grayscale palette for text and surfaces
- * Semantic Colors: `constructive` (green tones), `destructive` (red tones), `warning` (amber tones)

Typography:

- * Primary Typeface: Inter (fallback to system fonts for performance)
- * Hierarchy through size, weight, and letter-spacing
- * Base weight: 400, Bold weight: 600-700 for emphasis

Component Library (“BaseCore”):

- * React-based component library with TypeScript
- * Semantic prop system for maintainability

- * Constraint-based design to minimize design debt
- * Support for light/dark modes
- * Accessibility (WCAG 2.1 AA compliance)

Layout & Spacing:

- * Base spacing unit: 4px (following 4px grid system)
- * Max content width: 1200px centered
- * Consistent border radii: 6px (small), 10px (medium), 12px (large), 16px (cards)
- * 1px hairline borders for elevation instead of heavy shadows

Interaction Patterns:

- * Response time targets: 0.5-1s for most actions with visual feedback
- * Drag-and-drop with visual affordances (shadows, opacity, positioning feedback)
- * Contextual modals and side panels to maintain workflow context
- * Empty states with clear calls-to-action
- * Progressive onboarding with step indicators

Module-Specific Patterns:

- * **Project Management:** Kanban boards with Trello-inspired drag-and-drop
- * **CRM Plus:** Attio-inspired data grids with advanced filtering
- * **Accounting & Invoicing:** Ramp-inspired editorial minimalism with financial action highlights
- * **All Modules:** Monday.com-inspired component consistency and onboarding flows
- * **Dashboard:** QuickBooks-inspired customizable widget system

7. Implementation Technical Stack

- **Frontend Framework:** Next.js 14+ (App Router), React, Tailwind CSS, Lucide Icons.
- **Backend Framework:** FastAPI (Python 3.11+) asynchronous micro-services.
- **Database & Cache:** PostgreSQL with Row-Level Security (RLS) + Redis for session caching.
- **Billing Gateway:** Stripe API (SetupIntents, Subscriptions, Webhooks).
- **AI Orchestration:** Abacus.ai RouteLLM APIs with domain-specific system prompts.
- **Hosting & Edge Infrastructure:** Abacus.ai Cloud hosting with automated SSL termination.